

Geometric Structure in Neural Population Codes

A tool-comparison framework for causal neural geometry

Exploratory results on Steinmetz 2019

Elliott Tower

2026

The question

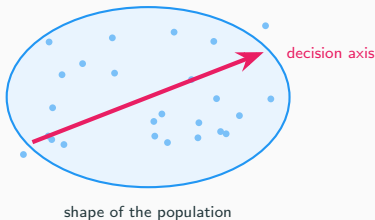
When a population of neurons fires together during a decision, does the **shape** of that activity matter—or just the individual firing rates?

Traditional view



VS.

Geometric view



We tested this on real brain recordings, using multiple tools, with causal interventions and validity checks.

What is “neural geometry”?

Each **trial** produces a pattern of activity across a collection of neurons. Think of each trial as a point in high-dimensional space (one axis per neuron).

“Geometry” = what **shape** do those points form?

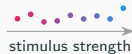
- Cluster into groups? (categories)
- Spread along a line? (gradient)
- Form a loop? (periodic variable)
- Lie on a flat sheet or curved surface?

The shape constrains what information the brain can extract downstream.

Clusters



Gradient



Loop

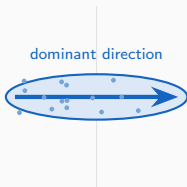


Flat



Two tools for measuring geometry

CKA (linear)

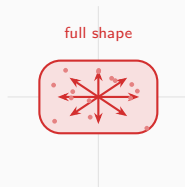


Compares the **main axis** of variation.

“Do the data stretch the same way?”

Best for low-dimensional data.

Procrustes (manifold)



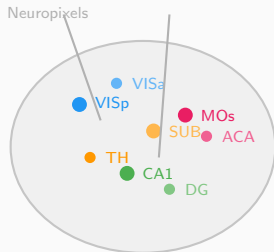
Compares the **full shape** via rotation + scaling.

“Do the data form the same cloud?”

Best for high-dimensional data.

Most prior work treats these as interchangeable. **They are not.**

The data: Steinmetz et al. 2019



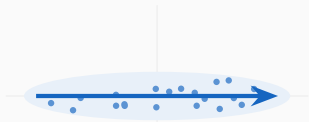
- **39 sessions** from **10 mice**
- **73 brain regions** recorded simultaneously
- Neuropixels probes ($\sim 30,000$ neurons total; ~ 50 – 500 per region per session)
- **Task:** visual discrimination
Two screens show visual noise at different contrasts. Mouse turns a wheel toward the higher-contrast side.
- 1,316 pairwise region comparisons

Steinmetz et al., Nature 2019

What we found: the tools disagree

Why CKA and Procrustes give opposite answers

Low-dimensional region (VlSp-type)



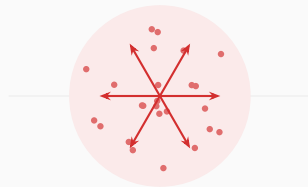
One direction dominates

CKA picks it up ✓

Procrustes sees the shape ✓

Both tools agree

High-dimensional region (MOs-type)



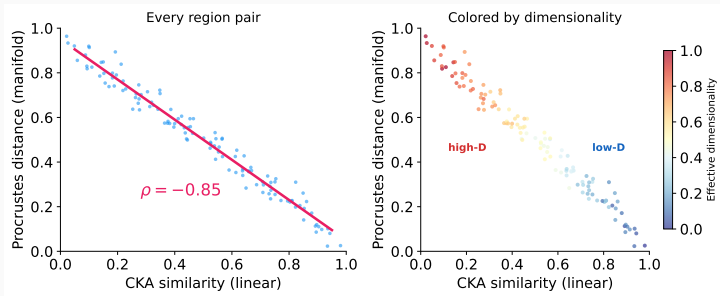
Activity spreads in many directions

CKA can't find a main axis → sees noise

Procrustes matches cloud shape → stable

Tools disagree

The data: CKA and Procrustes are anti-correlated

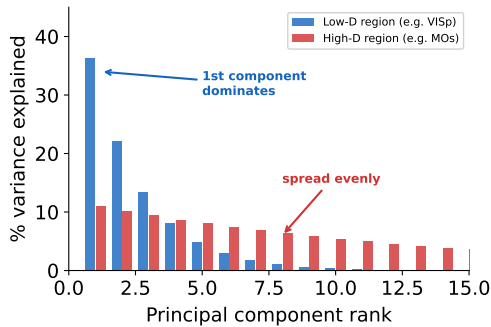


Each dot is a pair of brain regions. The two tools rank them in **opposite order** (correlation = -0.85 ; -1 would be perfect reversal).

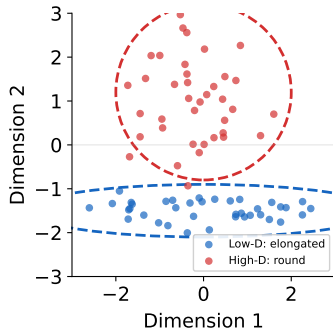
Why? The right panel colors dots by how “spread out” the neural activity is. Blue dots (compact activity) make both tools agree. Red dots (diffuse activity) make them disagree.

Once you account for this

What determines dimensionality?



How variance is distributed

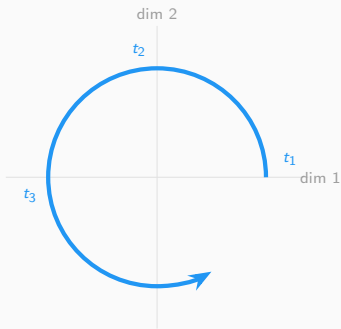


What this looks like in the data

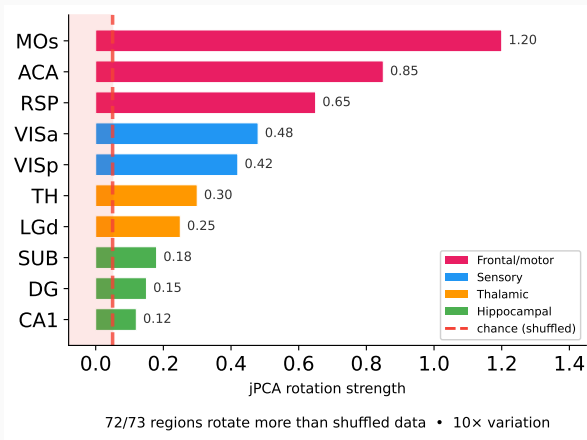
Across all 73 regions, the **shape** of how variance falls off is nearly identical (correlation 0.94 out of 1.0). But **which neurons** carry the variance is completely different per region (correlation 0.09, barely above zero). Same scaffolding, different content.

Every brain region rotates (but at very different speeds)

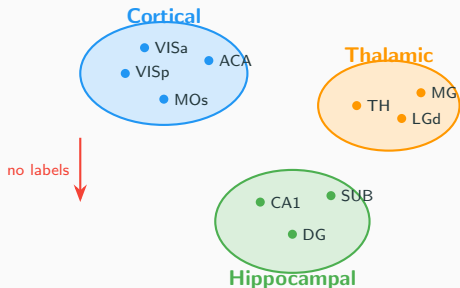
What rotation means



Neural activity traces out a circular path over time



Sanity check: geometry tracks known anatomy



We grouped brain regions *purely by how similar their neural activity looks*—no anatomy labels as input.

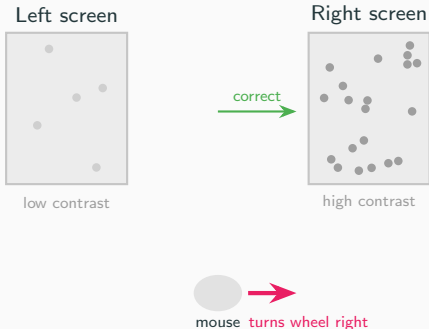
Result: the algorithm found three groups matching the textbook cortical / thalamic / hippocampal division on its own.

Why this matters: CKA is a general-purpose metric, not designed for neuroscience. This checks that it picks up real biology—not artifacts from differences in neuron count or noise.

Same result within individual sessions and across animals.

Is the geometry causal?

Recap: what the mouse is doing



The mouse sees two screens with random visual noise. One side has **higher contrast** (more visible dots). The mouse turns a wheel toward that side to get a reward.

“Evidence” = which side has higher contrast. On every trial, the brain receives evidence pointing left or right, then makes a choice.

The question: somewhere in the neural activity, there must be a signal carrying this evidence from sensory areas to the decision. We call it the “evidence subspace.”

Testing causality: swap the evidence signal between trials

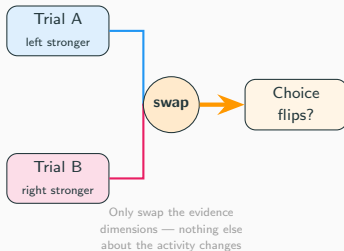
Showing that neural activity *correlates* with a decision is easy. Showing it *causes* the decision is hard.

Our test: take two trials where the mouse saw opposite evidence:

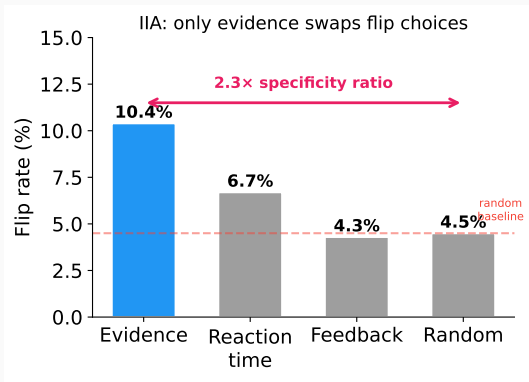
- Trial A: stronger on the **left** → mouse chose left
- Trial B: stronger on the **right** → mouse chose right

Now **swap only the evidence dimensions** of their neural activity (leaving everything else untouched) and re-run the choice decoder.

If the predicted choice flips, those dimensions



Result: swapping evidence dimensions flips choices



What each bar means:

- **Evidence**: swap the subspace that encodes which side has higher contrast
- **RT**: swap the subspace encoding reaction time
- **Feedback**: swap correct/error encoding
- **Random**: swap a random subspace

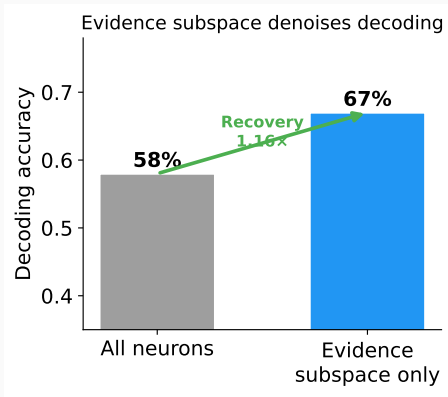
Key numbers:

Evidence flip rate: **10.4%**

Random baseline: **4.5%**

Ratio: **2.3 $\times$** ($p < 0.001$)

The evidence subspace is sufficient (and denoises)



If we restrict a classifier to *only* the evidence subspace dimensions, it decodes choice **better** than using all neurons.

Why? The evidence subspace acts as a denoiser—it strips away irrelevant neural variation that confuses the classifier.

20 of 24 regions pass the ≥ 0.70 recovery threshold.

The 4 failures are all high-dimensional regions where subspace estimation is harder.

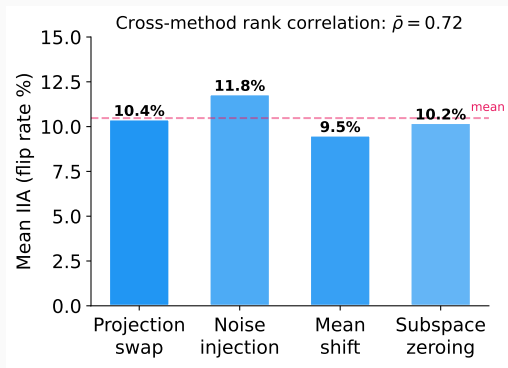
Together with IIA (necessity), this establishes the evidence subspace as a **causal mediator** of choice

Four different intervention methods agree

We tested **four different ways** to intervene:

1. Projection swap (replace component)
2. Noise injection (add Gaussian noise)
3. Mean-shift (shift toward opposite class)
4. Zeroing (remove component entirely)

These are mathematically different operations.
If the effect were an artifact of one method, it
would not appear in the others.



Regions that show high IIA with one method
show high IIA with all four. The effect is
method-invariant.

Did we fool ourselves?

What could go wrong?

Before trusting these results, we need to rule out simpler explanations:

Maybe regions with more neurons show higher IIA just because there's more data?

Maybe higher firing rates make it easier to detect any effect?

Maybe one unusual mouse drives the whole result?

Maybe temporal autocorrelation creates spurious signal?

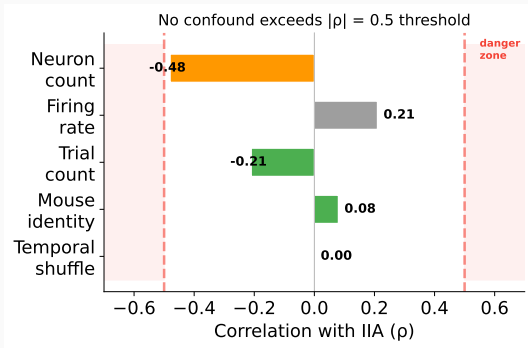
Maybe a random subspace would work just as well?

Maybe one intervention method is doing all the work?

Maybe the effect only appears with certain statistical tests?

We tested each of these. The framework comes from philosophy of science: validity criteria from Craver (2007), Shallice (1988), and Mayo (2018).

Confound control: what doesn't explain the effect



(ρ = correlation: 0 means no relationship, ± 1 means perfect; $|\rho| > 0.5$ is our danger zone.)

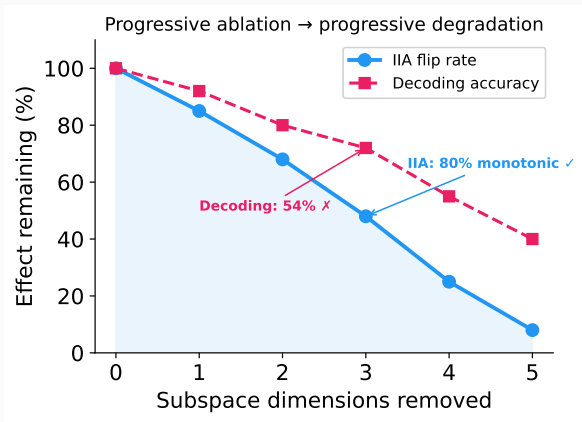
Clear passes:

- Temporal shuffle: $\rho = 0.001$
(scrambling time within trials doesn't change the effect — it's not a timing artifact)
- Mouse identity: $\rho = 0.08$
(no single mouse drives the result)
- Trial count: $\rho = -0.21$
- Firing rate: $\rho = 0.21$

Near-miss:

- Neuron count: $\rho = -0.48$

Graded response: remove dimensions one at a time



IIA (flip rate): monotonic dose-response in **80%** of regions ✓

Decoding accuracy: monotonic in only **54%** of regions ✗

The IIA (causal) measure shows clean dose-response. The decoding (correlational) measure is less clean—some regions have redundant coding where removing one dimension shifts load to others.

Mixed: IIA passes, ablation fails. The causal effect is graded; the correlational proxy is not.

Novel predictions: what geometry tells us that we didn't know

Geometric type predicts decoder choice

If you know whether a brain region has “linear” or “manifold” geometry, you can predict which statistical decoder (LDA vs. kNN) will work better—without ever training on decoding data.

correlation = -0.51 , highly significant
($p = 0.0002$, 50 regions)

Causal abstraction

Regions with stronger low-dimensional structure route evidence to choice more efficiently (moderate effect).

correlation = 0.29 , significant ($p = 0.014$)

Allen Atlas: an honest null

We tested whether anatomical wiring (from the Allen Mouse Brain Atlas) predicts which regions show strong causal evidence routing.
correlation = 0.03 , not significant ($p = 0.65$)

It doesn't. Physical wiring between brain regions does not predict which ones causally carry evidence. The geometry we found captures something that pure anatomy misses.

We report this null alongside the positive results. Three predictions made: two confirmed, one null.

Criteria adapted from Craver (2007), Mayo (2018), and Shallice (1988). We defined these before running the experiments, not after seeing the results.

What this means

Summary: what we found and what it means

Descriptive findings:

1. Linear and manifold tools anti-correlate (-0.85); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Rotation everywhere ($72/73$), but $10\times$ variation by region type
4. Geometry clusters match anatomy (sanity check)

Causal findings:

5. Evidence subspace causally mediates choice (IIA, 10.4% vs 4.5% baseline)
6. Effect is specific to evidence (not RT, feedback, random)
7. Effect holds across 4 intervention methods (avg. correlation 0.72)
8. Evidence subspace denoises decoding (recovery 1.16)

The punchline: Neural geometry is not metric-neutral—the tool you choose determines what you find, and dimensionality tells you which tool to trust. The causal structure we identify is modest but robust and specific.

Honest limitations

What limits these conclusions:

- **One dataset** (Steinmetz 2019)—needs replication
- **Binary task only**—unclear if geometry generalizes to richer tasks
- **Modest effect sizes**: 10.4% flip rate (baseline 4.5%)
- **Neuron count near-miss**: correlation -0.48
- **Double dissociation partial**: only 18% rate
- **Exploratory**: no preregistration

What we're doing about it:

- IBL cross-dataset replication (in progress)
- Zatka-Haas optogenetic validation (silencing data downloaded)
- Multi-task generalization
- Dimensionality correction for neuron count
- Preregistered hypotheses for replication

We think reporting limitations this explicitly is itself unusual in the field, and worth doing.

Intellectual grounding

This work draws on ideas from three research traditions:

- **Structured representation learning** (Esmaeili et al. 2019; Oppen et al. 2023; ICML 2025)—explicit structural priors improve identifiability of learned representations. Our dimensionality-based metric taxonomy is an inductive bias argument: the right prior should reflect the data's geometric structure.
- **Relational causal models** (Maier et al. 2013; Arbour et al. 2016)—brain regions interact as a network, not as independent samples. Standard causal inference tools need adaptation for this.
- **Dynamical systems in neuroscience** (Sussillo & Barak 2013; Vyas et al. 2020)—the connection between network dynamics and computational capacity. Our rotation findings connect to this tradition.

Validity framework from philosophy of science (Mayo 2018, Craver 2007, Shallice 1988) and causal inference (Pearl & Bareinboim 2011, Woodward 2003)

Code & results: [ZENODO DOI / GITHUB URL]

Neural geometry is not metric-neutral.

The tool determines the finding.

Dimensionality tells you which tool to trust.